

Biology experiments By Vasumitra Gajbhiye

1. Effect of surface area on diffusion

Method

- Use agar cubes that contain universal indicator.
- Cut two cubes of agar jelly, with sides of 1 cm.
- Cut one of the cubes into four smaller cubes, each with sides of 0.5 cm.
- Put the large cube into a test-tube. Put the four small cubes into the other test-tube.
- Add equal volumes of dilute hydrochloric acid to each tube, making sure that all of the cubes are covered with acid.
- Start the stopwatch. Time how many seconds it takes for the cubes in each test-tube to become fully red.

Safety

- Take care with the acid. Wear safety glasses in case of splashes.
- Take care with the sharp blade. Cut with the blade facing away from you. Place the jelly on a firm surface before you start to cut it.

Constant variable

- Concentration of acid
 - Temperature
 - Type of agar
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2. Effect of temperature on diffusion

Method

- Use agar cubes that contain universal indicator.
- Use at least 2 different temperatures.

- Cut two cubes of agar jelly, with sides of 1 cm.
- Add some hydrochloric acid in a test tube.
- Heat the test tube in water bath at known temperature.
- After equilibrating the temperature add one of the cube in it.
- Start the stopwatch. Time how many seconds it takes for the cubes in each test-tube to become fully red.
- Repeat for other temperatures.

Safety/ Precaution

- Take care with the acid. Wear safety glasses in case of splashes.
- Take care with the sharp blade. Cut with the blade facing away from you. Place the jelly on a firm surface before you start to cut it.
- Use tongs to handle hot test tubes.

Constant variable

- Surface area of cubes
- Concentration of acid
- Type of agar

3. Effect of concentration gradient on diffusion

Method

- Use hydrochloric acids of at least 3 different concentrations.
- Use agar cubes that contain universal indicator.
- Cut three cubes of agar jelly, with sides of 1 cm.
- Add some hydrochloric acid of known concentration in a test tube.
- Heat the test tube in water bath at known temperature.
- After equilibrating the temperature add one of the cube in it.
- Start the stopwatch. Time how many seconds it takes for the cubes in each test-tube to become fully red.

- Repeat for other concentrations of hydrochloric acid.

Safety/ Precaution

- Take care with the acid. Wear safety glasses in case of splashes.
- Take care with the sharp blade. Cut with the blade facing away from you. Place the jelly on a firm surface before you start to cut it.
- Use tongs to handle hot test tubes.

Constant variable

- Surface area of cubes
 - Temperature of Acid
 - Type of agar
-

4. Effect of concentration on Osmosis

Method

- Use at least 2 different concentrations of sugar solution
- Make different concentrations of sugar solution by adding different mass of sugar into same volume of water.
- Collect a piece of dialysis tubing. Tie one end of it around itself in a tight knot.
- Use a dropper pipette to carefully add **known volume** of the most concentrated sugar solution in the dialysis tubing (**V1**).
- Use some strong thread to tie the other end of dialysis tubing very tightly.
- Place the dialysis tubing inside a beaker of water.
- Start the timer. Wait for 5 minutes.
- Remove the dialysis tubing. Empty it's content in a measuring cylinder and measure the volume of liquid in it as V2.
- Calculate change in volume = $V2 - V1$.
- Repeat the experiment using different concentration of sugar solution in the dialysis tubing.

Safety/ Precaution

- Rinse the outside of dialysis tubing after filling it with sugar solution
- Use a new clean dropper pipette to fill dialysis tubing with sugar solution.

5. Effect of different concentrations on Potato cylinders

Method

- Peel the potato. Use a cork borer or knife to cut six cylinders (or cuboids) from the potato. Cut the ends from each cylinder so that each one is the same length – 40 mm is a good length to use.
- Use at least 2 different concentrations of sugar solution.
- Make different concentrations of sugar solution by adding different mass of sugar into same volume of water.
- Fill a beaker with one sugar solution and add two potato cylinders in it.
- Leave the potato strips in their liquids for about 20 minutes.
- After 20 minutes, take the potato strips out of their solutions. Measure the length of each one again. Calculate the change in length of each strip.
- Repeat using different concentration of sugar solution.
- Repeat using distilled water.

Safety/ Precaution

- Use the cork bore or knife carefully. cut away from the body on a wooden board or flat tile.

Constant Variable

- Initial length of potato cylinder (or cuboid)
 - Same type of potato
 - Same temperature
 - Same soaking time
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Testing a leaf for starch

Method

- Boil some water in a beaker. Place the leaf in it for 30seconds.
- Turn out the flame.
- Put some alcohol into a boiling tube. Stand the tube of alcohol in the hot water.
- Use the forceps to remove the softened leaf from the hot water. Drop it into the tube of alcohol.
- Leave the leaf in the alcohol until all the chlorophyll has come out of the leaf.
- The leaf will now be brittle (it will break easily). Carefully remove it from the alcohol, and dip it into hot water again to soften it.
- Spread out the leaf on a white tile and cover it with iodine solution. A blue-black colour shows that the leaf contains starch.

Safety/ Precaution

- Wear gloves and lab coat
- Alcohol is flammable so do not heat it in Bunsen burner.
- Handle hot test tubes with tongs
- use forceps to handle the fragile leaves.

Constant Variable

- No constant Variable for this experiment

Note

- This is a very important experiment as it serves as a foundation for further investigations. So make sure you understand it completely.

Investigating the necessity for chlorophyll for photosynthesis

Method

- In this investigation you will need a

variegated leaf. A variegated leaf is one that has some white areas (with no chlorophyll) and some green areas (with chlorophyll)



- Perform the starch test on this leaf
- You will notice that only the area that is green will turn blue-black. The white area will remain yellow-brown
- This proves that chlorophyll is necessary for photosynthesis as no starch is produced in the region where there is no chlorophyll.

Safety/ Precautions

- Same safety/ precautions as for test for starch

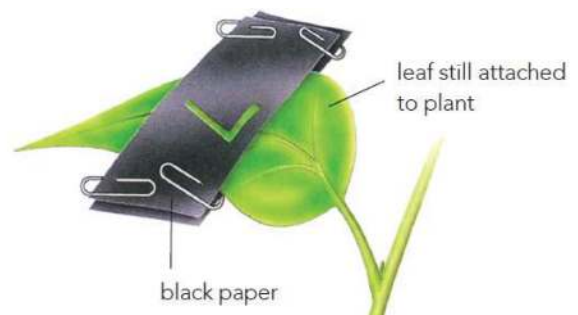
Control Variable

- No control variable for this experiment.

Investigating the necessity for light for photosynthesis

Method

- Cut out a shape from the black paper, which you can attach to a leaf. Like the one in this diagram.
- Place a potted plant in a very dark room for 3 days to de-starch it.
- Attach the paper to a leaf on the potted plant. Do not take the leaf off the plant!
- Put the plant into a place where it gets plenty of light. Leave it until the next day.
- Now remove the leaf from the plant. Remove the black paper from the leaf.
- Test the leaf for starch.



- You will observe that the part that is covered by the black paper shows negative result for starch test and the uncovered part shows positive test.

Safety/ Precautions

- Same safety/ precautions as for test for starch

Control Variable

- No control variable for this experiment.

Investigating the necessity for carbon dioxide for photosynthesis

Method

- Place two potted plants inside bell jars. Place them in a dark room for 2 days to de-starch.
- Then place a beaker full of potassium hydroxide in one jar and a beaker full of water in another jar.



- Place the apparatus in a light place and leave it for at least one day.
- Remove one leaf from both the plants and test both leaves for starch.
- You'll observe that leaf from potassium hydroxide plant remain yellow-brown while leaf from water plant turn to blue black. Showing that carbon dioxide is necessary for photosynthesis.

Safety/ Precaution

- Wear gloves while handling potassium hydroxide.
- Wear safety goggles.

Control Variables

- Species of plant
- Age of plant
- Size of plant
- Duration of de-starching
- Same light intensity

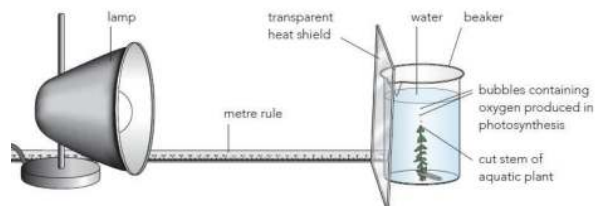
Note

- The potassium hydroxide is used to absorb carbon dioxide from the bell jar. So that there is absence of carbon dioxide in one plant.

Investigating the effect of varying light intensity on the rate of photosynthesis

Method

- Collect a piece of aquatic plant. Cut off a piece about 7 to 9 cm long **under the water**.



- Fill a beaker with **tap** water. place the aquatic plant in the beaker upside down, i.e. the cut side should be upwards.
- Place lamp 10 cm from the beaker. Place a transparent heat shield between the lamp and beaker. If transparent heat shield is not available use a measuring cylinder or beaker full of water and place it between the plant containing beaker and lamp.
- Turn on the lamp. Wait for some time so that the plant adapt to the changed environment.
- Start the stopwatch or timer and count the number of bubbles released from the cut end of the stem in one minute. Record this, and then repeat two more times.

- Move the lamp 10cm further from its initial position. Measure the new distance, and repeat the investigation.

Safety/ Precaution

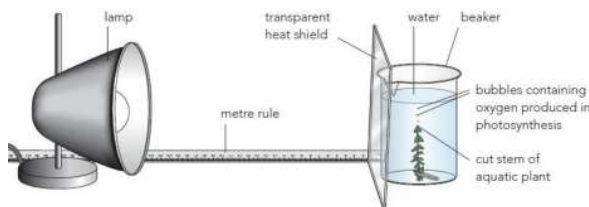
- Cut the plant under the water
- Do not handle the lamp with wet hands.

Control Variables

- Species of plant
- Age of plant
- Size of plant
- Same time for counting the number of bubbles.

Investigating the effect of varying carbon dioxide concentration on the rate of photosynthesis

- Collect a piece of aquatic plant. Cut off a piece about 7 to 9 cm long **under the water**.



- Fill a beaker with **distilled** water. Add known mass of sodium hydrogen-carbonate to the water and stir. place the aquatic plant in the beaker upside down, i.e. the cut side should be upwards.
- Place lamp 10 cm from the beaker. Place a transparent heat shield between the lamp and beaker. If transparent heat shield is not available use a measuring cylinder or beaker full of water and place it between the plant containing beaker and lamp.
- Turn on the lamp. Wait for some time so that the plant adapt to the changed environment.
- Start the stopwatch or timer and count the number of bubbles released from the cut end of the stem in one minute. Record this, and then repeat two more times.

- Repeat the investigation with different mass of sodium hydrogen-carbonate in water.
- Repeat the investigation using only distilled water.

Safety/ Precaution

- Cut the plant under the water
- Do not handle the lamp with wet hands.

Control Variables

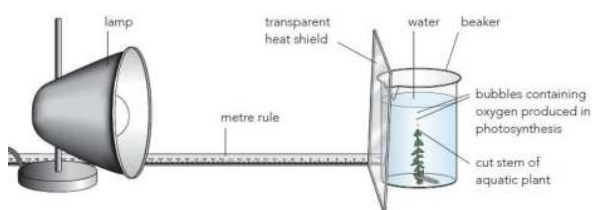
- Species of plant
- Age of plant
- Size of plant
- Same time for counting the number of bubbles.
- Same light intensity.

Note

- The sodium hydrogen-carbonate is a source of carbon dioxide for the plant.

Investigating the effect of varying temperature on the rate of photosynthesis

- Collect a piece of aquatic plant. Cut off a piece about 7 to 9 cm long **under the water**.



- Fill a beaker with **tap** water. Place this beaker in a larger beaker that is a thermostatically controlled water bath at known
- Place the aquatic plant in the beaker upside down, i.e. the cut side should be upwards.
- Place lamp 10 cm from the beaker. Place a transparent heat shield between the lamp and beaker. If transparent heat shield is not available use a measuring

cylinder or beaker full of water and place it between the plant containing beaker and lamp.

- Turn on the lamp. Wait for some time so that the plant adapt to the changed environment.
- Start the stopwatch or timer and count the number of bubbles released from the cut end of the stem in one minute. Record this, and then repeat two more times.
- Repeat the investigation for different temperatures.

Safety/ Precaution

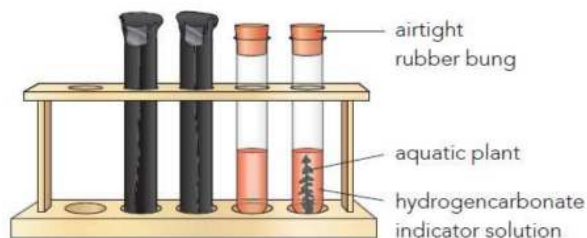
- Cut the plant under the water
- Do not handle the lamp with wet hands.

Control Variables

- Species of plant
- Age of plant
- Size of plant
- Same time for counting the number of bubbles.

Investigating the effect of light and darkness on gas exchange in an aquatic plant

- Pour same volume of hydrogen-carbonate indicator into four tubes.
- Record the colour of the indicator in each tube.



- Place a piece of aquatic plant in the indicator in **two** tubes.
- Use black paper to wrap around one of the tubes with a plant in it, and one without.
- Stand all four tubes in the light and leave them for at least two hour.

- When the indicator has changed colour in at least one of the tubes without a covering, remove the black paper from the two covered tubes. Record the colour of the indicator in each tube.

Safety/ Precaution

- When covering the test tube with black paper make sure the tubes are completely covered.

Control Variables

- Species of plant
- Age of plant
- Size of plant
- Same volume of hydrogen-carbonate indicator

Note

- Colour of hydrogen-carbonate indicator
 - Low concentration → purple
 - Medium concentration (same as clean dry air) → red
 - High concentration → yellow
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Investigating the effect of temperature on pathway of water through the above-ground parts of a plant

Method

- Add known volume of water in a beaker. Add known volume of dye in the beaker and stir. Place the beaker in thermostatically controlled water bath at set temperature.
- Equilibrate the temperature then place a celery stalk in the container for at least 2 hours.
- Remove the stalk.
- From the bottom part of the stalk start cutting 5mm sections of the stalk using a small rule and knife.

- Keep cutting until you stop seeing the colour of dye in the xylem of the cut section.
- Count the number of sections that show the coloured dye in stalk. Multiply the number of sections by 5 to get the approximate distance travelled by the dye.
- Repeat using different temperatures.

Safety/ Precautions

- Take care with the sharp knife or scalpel. Cut away from your body towards the wooden board or tile.
- Try not to get the dye on your skin or clothes, as it will be very difficult to wash off.

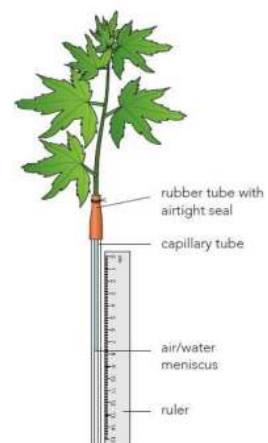
Control Variable

- Same type of celery stalk
- Same diameter of stalk
- Same volume of water and same volume of dye to maintain same concentration of dye.

Using a potometer to investigate the effects of wind speed on the rate of transpiration

Method

- Fill a potometer with water.
- Cut a plant shoot with knife and attach it to the potometer.
- Perform both of the steps mentioned above under water.
- Remove the potometer from water and make sure there are no air bubbles. If there are you might have to perform the whole process again.
- Attach a scale to the potometer so that you can measure the distance moved by water.
- Keep a fan at set distance from the potometer. Turn it on.
- Leave the apparatus in a light.



- As the plant transpires, the water it loses is replaced by water taken up through the cut end of the stem. Air will be drawn in at the end of the capillary tube.
- Start the stop watch. Measure the distance moved by the air meniscus in 2 hours.
- Repeat using different intensity of fan speed.

Safety/ Precautions

- Do not perform the experiment until there are no water bubbles in the tube.
- When using knife cut away from the body.

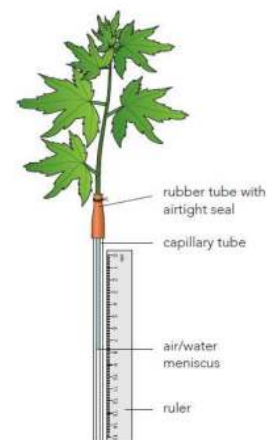
Control Variables

- Species of plant
- Age of plant
- Size of plant
- Room temperature
- Room humidity
- Distance between lamp and potometer

Using a potometer to investigate the effects of temperature on the rate of transpiration

Method

- Fill a potometer with water.
- Cut a plant shoot with knife and attach it to the potometer.
- Perform both of the steps mentioned above under water.
- Remove the potometer from water and make sure there are no air bubbles. If there are you might have to perform the whole process again.
- Attach a scale to the potometer so that you can measure the distance moved by water.



- Keep it in refrigerator at known temperature with a light source at known distance to maintain cool temperature. Keep it in incubator at known temperature with a light source at known distance to maintain hot temperature.
- As the plant transpires, the water it loses is replaced by water taken up through the cut end of the stem. Air will be drawn in at the end of the capillary tube.
- Start the stop watch. Measure the distance moved by the air meniscus in 2 hours.
- Repeat using different temperatures.

Safety/ Precautions

- Do not perform the experiment until there are no water bubbles in the tube.
- When using knife cut away from the body.

Control Variables

- Species of plant
- Age of plant
- Size of plant
- Wind speed
- Humidity

Investigating the effect of physical activity on heart rate

Method

- You'll need one partner for the experiment.
- Measure the initial pulse rate of your partner.
- Ask them to run 100m And then measure their heart rate.
- Let them relax completely and let their heart rate return to normal.
- Repeat for different distances to increase or decrease physical activity intensity.

Safety/ Precaution

- Do not force your partner to push beyond their physical limit

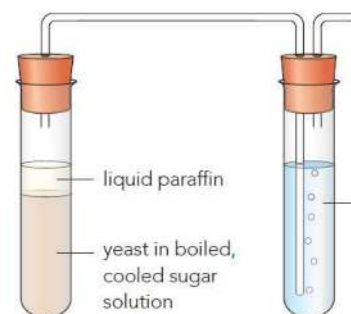
Constant variable

- Running speed
 - Gender
 - Age
 - Fitness
-

Investigating the effect of temperature on respiration in yeast

Method

- Add known volume of yeast suspension in a test tube.
- Add known volume of glucose in it.
- Connect a bung. Connect a delivery tube to the other test tube as shown in the diagram.
- Place the yeast test tube in a thermostatically controlled water bath at known temperature.



- Equilibrate the temperature and then count the number of bubbles released for 360s.
- Repeat and average the number of bubbles released at that temperature.
- Repeat for other temperatures.

Safety/ Precautions

- Handle hot test tubes with tongs.

Constant variable

- Volume of yeast suspension
 - Volume of glucose
 - Duration of experiment.
-

Mini Experiments

- Test for Reducing Sugar.
 - Add benedict reagent
 - Heat it in water bath at 85°C
 - Colour change from blue to green/yellow/orange/red
- Test for Protein.
 - Add Biuret reagent
 - Colour change from Blue to Purple
- Test for Vitamin C.
 - Add DCPIP solution
 - Solution decolourise, i.e. change colour from blue to colorless
- Test for Fat.
 - Mix sample with ethanol.
 - Add this solution to water.
 - Positive test if emulsion is seen, i.e. the solution turn milky-cloudy.

- Test for Starch.
 - Add Iodine **solution**
 - Colour change from yellow-brown to blue-black